Comparative Eco-Physiological and Biomechanical Survivability: A Multi-Disciplinary Analysis of Pleistocene Hominins

Introduction: The Calculus of Pleistocene Survivability

The evolutionary trajectory of the genus *Homo* throughout the Middle and Late Pleistocene represents a highly complex theatre of biological adaptation, ecological competition, and extreme environmental volatility. For generations, paleoanthropologists and evolutionary biologists have interrogated the extinction of archaic hominins—most notably the Neanderthals (*Homo neanderthalensis*) and the Denisovans—and the concurrent, unprecedented proliferation of anatomically modern humans (*Homo sapiens*). Historically, the displacement of archaic populations was frequently attributed to a singular, overriding advantage possessed by modern humans, often theorized as vastly superior cognitive capacity, advanced linguistic abilities, or an intrinsic propensity for systemic violence. However, contemporary paleoanthropological discoveries, advanced genomic sequencing, and high-fidelity biomechanical modeling have dismantled these monolithic theories, revealing a significantly more nuanced paradigm of survivability.

The survival of *H. sapiens* through dramatic climatic shifts, ecological turnover, and natural disasters, juxtaposed against the decline of their robust cousins, was dictated by a multifaceted intersection of skeletal morphology, locomotor energetics, integumentary physiology, and socio-demographic resilience.[1] Survivability in the Pleistocene was not merely a function of raw physical strength or extreme morphological specialization; rather, it was a calculus of energetic efficiency, biomechanical versatility, and the capacity to buffer environmental shocks through expansive socio-cultural safety nets.[1]

This comprehensive analysis systematically deconstructs the comparative survivability of these hominin lineages across several critical physiological and behavioral domains. It examines fundamental divergences in skeletal robusticity, encompassing bone density, buoyancy dynamics, and the specific morphological adaptations that facilitated both aquatic foraging and sophisticated upper-limb projectile biomechanics. Furthermore, it explores vital physiological adaptations to severe abiotic stressors, namely intense solar radiation and high-altitude hypoxia, detailing how the introgression of archaic alleles into the modern human genome catalyzed rapid adaptation.[3] The report subsequently delineates the profound disparities in metabolic energetics, thermoregulatory demands, and caloric economics that shaped the daily subsistence strategies of these species.[5] Finally, it addresses the ultimate determinant of long-term disaster survivability: the architecture of social networks, outbreeding practices, and demographic connectivity.[1] Through a synthesis of recent archaeological evidence, genomic data, and digital ecological modeling, a unified framework emerges, demonstrating that the ascendancy of modern humans was predicated on an evolutionary trend toward profound systemic flexibility.

Osteological Architecture and the Genetic Basis of Gracility

The evolutionary divergence between modern humans and archaic hominins is perhaps most starkly illustrated by fundamental variations in overall skeletal robusticity, specific trabecular bone density, and the consequential impacts on physical maneuverability and metabolic maintenance. A defining characteristic of the modern human skeleton is its relative gracility—a condition characterized by lighter bone mass relative to overall body size.[8] For decades, this gracilization was interpreted primarily as a late-stage epigenetic response to reduced mechanical loading, a byproduct of the transition toward increased sedentism and agricultural practices during the Holocene.[8]

However, recent high-resolution genomic and morphological analyses have identified a distinct genetic basis for this divergence, indicating that the reduced bone robustness of modern humans was actively mediated by specific evolutionary pathways. Research has isolated variants of the *LRP5* gene that significantly contributed to the reduced bone density observed in modern *H. sapiens* when compared to the highly robust, thick-boned skeletons of Neanderthals and Denisovans.[9] This genetic mechanism suggests a fundamental divergence in bone architecture that predates the agricultural revolution, framing gracility not merely as a consequence of behavioral changes, but as an adaptive or pleiotropic genetic shift carrying significant implications for overall physiological economy.[9]

To quantify this morphological divergence across evolutionary time, researchers have utilized peripheral quantitative computed tomography and microtomography to measure the trabecular bone fraction (TBF) within the limb epiphyses of various hominin taxa spanning several million years.[8] The resulting data conclusively demonstrates that recent modern humans exhibit remarkably low trabecular density throughout their upper and lower limb joints. In stark contrast, extinct hominins—specifically Neanderthals and pre-Holocene early *H. sapiens*—retained the high levels of trabecular density that were characteristic of earlier hominins and nonhuman primates.[8]

Anatomical Element	**Taxon**	**Mean Trabecular Bone Fraction (TBF)**	**Standard Deviation**	**TBF Range**
Proximal Humerus	Recent *Homo sapiens*	0.20	0.02	0.16–0.26
Proximal Humerus	*Homo neanderthalen*	0.27	N/A	0.24–0.29

	sis			
Proximal Humerus	Early *Homo sapiens*	0.26	N/A	N/A
Proximal Ulna	Recent *Homo sapiens*	0.25	0.05	0.16–0.39
Proximal Ulna	*Homo neanderthalensis*	0.48	N/A	N/A
Proximal Ulna	Early *Homo sapiens*	0.33	N/A	N/A
Distal Radius	Recent *Homo sapiens*	0.21	0.03	0.16–0.27
Distal Radius	*Homo neanderthalensis*	0.38	N/A	N/A
Distal Tibia	Recent *Homo sapiens*	0.23	0.04	0.16–0.32
Distal Tibia	*Homo neanderthalensis*	0.40	N/A	N/A
Distal Tibia	Early *Homo sapiens*	0.37	N/A	N/A

The data provided in the comparative analysis invalidates the hypothesis of a slow, consistent, and gradual shift toward lighter skeletons throughout the Pleistocene.[8] Instead, it confirms that high trabecular values, alongside variations in overall skeletal robusticity, persisted vigorously until a relatively recent, precipitous decline in modern humans.[8] Interestingly, while the upper limb elements do not reflect a temporal decrease in trabecular density throughout the Pleistocene, the distinctiveness of the low density in recent modern humans is highly marked in the lower limb joints.[8] This differential gracilization between the upper and lower appendicular skeleton heavily implies that, alongside the genetic influence of the *LRP5* gene, radical changes in locomotor mobility and the cessation of highly nomadic hunter-gatherer behaviors acted as primary behavioral drivers for lower-limb skeletal gracilization in the Holocene.[8]

Pachyosteosclerosis and the Archaic Aquatic Foraging Paradigm

The retention of extreme skeletal robusticity and high trabecular density in Neanderthals and *Homo erectus* points to a specific physiological baseline known in osteological literature as pachyosteosclerosis.[11] Pachyosteosclerosis is a condition characterized by a combination of unusually thick cortical bone (pachyostosis) and highly dense, compact internal bone architecture with remarkably narrow medullary canals (osteosclerosis).[11] While often viewed merely as a terrestrial adaptation for absorbing immense physical trauma or accommodating massive musculature, comparative mammalian biology offers a highly compelling alternative explanation rooted in aquatic exploitation.

Across the vertebrate spectrum, pachyosteosclerosis is an exceedingly rare condition in purely terrestrial mammals. It is, however, a hallmark adaptation among semi-aquatic and shallow-diving marine tetrapods, such as the Sirenia (manatees and dugongs) and early evolutionary forms of Cetacea and Pinnipedia (particularly the Phocidae, or true seals).[11] In these marine taxa, pachyosteosclerotic bone serves a critical mechanical function: it acts as hydrostatic ballast.[11] This hyper-dense skeletal architecture provides static buoyancy control, counteracting the natural positive buoyancy of the lungs and adipose tissue.[11] By neutralizing buoyancy, these animals can effortlessly sink and maintain their position in shallow waters, allowing them to wade and dive for slow-moving or sessile benthic foods—such as aquatic vegetation, hard-shelled mollusks, and crustaceans—without expending massive amounts of muscular energy to prevent themselves from floating to the surface.[11]

The presence of widespread pachyosteosclerotic traits in archaic *Homo* heavily implies that part-time shoreline collection of aquatic foods was a defining element of their ecological niche and survivability strategy.[11] Early Pleistocene *H. erectus*, which exhibited some of the most profound pachyosteosclerosis in the hominin lineage, frequently occupied habitats closely associated with large bodies of permanent water and water-dependent molluscan fauna.[11] While the pachyosteosclerosis observed in Neanderthals was roughly half as pronounced as that of *H. erectus*, it remained a defining feature of their heavy skeletons, particularly evident in the occipital region of the cranium.[16] This heavy skeletal ballast would have facilitated energy-efficient shallow diving and coastal wading.[16]

Further substantiating the archaic aquatic foraging paradigm is the exceptional prevalence of external auditory exostoses (EAE) within Neanderthal populations.[18] EAE, clinically referred to as "swimmer's ear" or "surfer's ear," manifests as dense, bony protrusions growing into the external auditory canal.[18] This pathology is a direct osteological response to repeated and prolonged exposure to cold, moist environments, serving as the body's mechanism to protect the delicate tympanic membrane from thermal shock.[18] While EAE occurs in approximately 20% to 25% of archaic and early modern human skulls, comprehensive analyses by paleoanthropologists have revealed that it is present in a staggering 56.5% of examined Neanderthal remains.[18] Nearly half of the 23 specifically examined Neanderthal crania exhibited mild to severe cases of this condition.[20]

In modern clinical studies, EAE is overwhelmingly correlated with communities residing between latitudes 30° and 45° north and south of the equator, specifically among individuals who engage in frequent aquatic activities in water temperatures below 19°C.[19] The extreme frequency of EAE in Neanderthals is matched in recent human history solely by indigenous populations operating in chronic, cold-water maritime climates.[19] Importantly, while severe EAE can cause unilateral conductive hearing loss in terrestrial environments, it does not impair bone-conduction hearing while submerged in water.[22] Therefore, EAE was not a debilitating terrestrial handicap for a Pleistocene hunter-gatherer, but rather an adaptive response to frequent aquatic resource exploitation.[22]

This persistent aquatic foraging provided Neanderthals with access to highly concentrated marine and riverine resources, such as shellfish, crayfish, and aquatic plants.[16] These littoral foods are remarkably rich in brain-specific nutrients—most notably docosahexaenoic acid (DHA), iodine, taurine, and essential oligo-elements.[22] The consumption of these critical nutrients was likely indispensable for supporting the massive metabolic and structural requirements of the Neanderthal brain, which averaged 1,640 cubic centimeters in males, surpassing the average volume of living human populations.[16] Thus, the heavy, pachyosteosclerotic skeleton of the Neanderthal was a finely tuned instrument for extracting vital neuro-nutritional resources from coastal and riverine ecosystems, enhancing their survivability during periods when terrestrial megafauna were scarce.[16]

Modern Human Buoyancy, Swimming Mechanics, and the Myth of Bone Density

While the heavy skeletal architecture of archaic hominins facilitated static buoyancy for shallow-water foraging, it simultaneously hindered their ability to engage in dynamic, high-speed surface swimming or to efficiently traverse deep, fast-moving bodies of water. The gracilization of the *H. sapiens* skeleton fundamentally altered hominin buoyancy dynamics, replacing the static ballast system of the Neanderthal with a dynamic buoyancy control system heavily influenced by soft tissue composition, lung volume, and biomechanical alignment.[14]

For decades, a pervasive and scientifically inaccurate narrative suggested that variations in bone density among different modern human populations strictly determined their capacity to float.[25] Specifically, a damaging misconception posited that individuals of African, Caribbean, and Asian heritage possessed "heavy bones" that rendered them less buoyant and less capable of surviving in deep water environments.[25] This myth has recently been unequivocally debunked by exhaustive anthropometric and biomechanical research. A landmark study conducted by the University of Portsmouth in partnership with the Black Swimming Association achieved a 93% success rate in floatation among participants of African, Caribbean, Asian, and mixed heritage.[25] Out of 96 participants, 89 were able to float competently for a sustained

duration of two minutes following proper biomechanical instruction.[25]
This research confirms that modern human buoyancy is not dictated by the fractional variations in skeletal density between populations, but rather by the intricate relationship between the body's center of gravity and its center of buoyancy (flotation).[25] In the human body, the center of buoyancy (located in the air-filled thoracic cavity) and the center of gravity (located lower in the pelvis) are naturally divergent.[26] When a human attempts a motionless float, this spatial divergence generates a rotational torque. Rather than floating in a perfectly horizontal, streamlined position, the human body naturally rotates until the centers of gravity and buoyancy align vertically.[26] The water supports the weight of the motionless swimmer precisely at this individualized angle.[26] The traditional assumption that there is only one "correct" way to float—the horizontal "starfish" position—failed to account for these natural biomechanical alignments.[25]
Furthermore, human buoyancy is profoundly modulated by adipose tissue distribution and muscular density. Somatotypic variations driven by climatic adaptations directly influence aquatic performance.[26] For instance, indigenous Arctic populations, such as the Inuit, exhibit a body morphology characterized by higher proportions of subcutaneous fat and a rounder body shape—adaptations evolved strictly to minimize heat loss in freezing terrestrial environments.[26] Fortuitously, this combination of high fat-to-muscle ratio and morphological roundness significantly decreases specific gravity, making them exceptionally naturally buoyant.[26] Conversely, populations adapted to high-heat environments with extreme leanness and high muscle-to-bone mass ratios may exhibit negative buoyancy (often colloquially termed "sinkers"), requiring continuous dynamic kinetic output to maintain surface positioning.[26]
The reduced trabecular density of the modern human skeleton ultimately decoupled *H. sapiens* from the bottom-dwelling constraints of pachyosteosclerosis.[8] This dynamic buoyancy granted modern humans unparalleled versatility in aquatic environments, enabling rapid surface swimming, the efficient crossing of treacherous riverine barriers during migrations, and the exploitation of deep-water resources that were physically inaccessible to the heavily ballasted Neanderthals. When confronted with rapid flooding, tsunamis, or the necessity to escape terrestrial predators by taking to deep water, the buoyant, gracile anatomy of *H. sapiens* provided a distinct, life-saving advantage over previous hominins.[25]

The Hominin Shoulder Complex: Clavicular Allometry and Glenoid Morphology

The evolutionary capacity to execute complex, high-velocity movements of the upper limbs constitutes a defining biomechanical divergence between Neanderthals and *H. sapiens*. The morphological architecture of the hominin shoulder complex—comprising the clavicle, the scapula, and the proximal humerus—dictates the available range of motion, the potential for kinetic force generation, and the relative stability of the glenohumeral joint under extreme stress.[27]
In the broader context of primate evolution, hominoids are distinguished by possessing longer clavicles relative to body mass when compared to typical quadrupedal monkeys.[29] This

elongation reflects a lateral reorientation of the hominoid glenoid cavity, shifting the shoulder joint laterally away from the ribcage to facilitate a broader range of motion.[29] Modern humans conform closely to this general hominoid distribution.[29] Neanderthals, however, alongside some Upper Paleolithic *H. sapiens*, exhibited clavicles that were exceptionally long and gracile, displaying marked positive allometry in clavicle length relative to their overall size.[29] When visually assessing a Neanderthal skeleton in dorsal view, their clavicular morphology reveals that the scapula was positioned higher on the thorax than in modern humans.[27] Despite this extreme absolute length, when scaled proportionally to their massive body masses and broad thoraxes, the clavicular lengths of Neanderthals are functionally indistinguishable from those of Late Pleistocene and recent modern humans.[30]

The critical biomechanical differentiator between the two species resides not in the clavicle, but in the intricate morphology of the scapula, specifically the shape and orientation of the glenoid fossa (the socket of the shoulder joint).[31] Neanderthal scapulae are characterized by a broad overall shape, but they possess a glenoid fossa that is distinctively long, narrow, and exceptionally flat across its articular surface.[31] Furthermore, their scapulae exhibit pronounced dorsal sulci on the axillary borders.[31]

In stark contrast, modern humans possess a glenoid fossa that is relatively wider in the antero-posterior axis and more concave in its articular curvature.[32] This morphological variance dictates the fundamental compromise inherent in all joint anatomy: the inverse relationship between stability and mobility.[34] A narrow, flat glenoid fossa, as seen in the Neanderthal, prioritizes extreme joint stability.[32] This architecture is designed to withstand massive compressive forces and facilitate highly powerful, controlled movements in the sagittal plane, such as the two-handed thrusting of heavy, close-quarters weaponry.[32] However, this stability inherently restricts the degree of extreme dorsoventral glenohumeral movement and complex circumduction.[32]

The wider, more concave glenoid cavity of modern *H. sapiens* sacrifices a degree of raw compressive stability in favor of an exceptionally heightened range of motion.[32] This enhanced mobility is an absolute anatomical prerequisite for the complex kinetic chain required in high-velocity overhand throwing.[28] During a throwing motion, the scapula must act as a stable base while simultaneously moving in three dimensions around the trunk to maintain glenohumeral alignment.[28] It must retract during the cocking phase, upwardly rotate to prevent subacromial impingement of the rotator cuff, and aggressively protract during acceleration and deceleration.[28] The modern human scapula is perfectly adapted to facilitate this highly dynamic, multi-axial choreography, granting *H. sapiens* an unprecedented ability to interact forcefully with their environment at a distance.[28]

Biomechanics of High-Velocity Projectiles Versus Ambush Power

The evolutionary refinement of the modern human shoulder complex effectively transformed the *H. sapiens* upper limb into a sophisticated biological projectile launcher. The mechanics of high-velocity throwing involve forces and rotational speeds unmatched by any other primate

species.[36]
When a modern human executes an overhand throw, they harness the elastic properties of their connective tissues.[36] During the late cocking phase of the throw, the arm reaches an extreme position of thoracohumeral abduction (approximately 93 degrees) and experiences a massive external rotation of the humerus (often exceeding 64 degrees).[38] This extreme rotation acts to heavily load the ligaments and tendons crossing the shoulder joint, stretching them analogous to the elastic bands of a slingshot.[36] During the subsequent acceleration phase, this stored elastic potential energy is released rapidly and forcefully, driving the internal rotation of the upper arm with extraordinary speed.[36] The forces generated during the early deceleration phase are tremendous, subjecting the joint to estimated compression forces of 1090 Newtons and anteroinferior shear forces of 130 Newtons.[38]
Chimpanzees, lacking this specific glenohumeral orientation and elastic energy storage capacity, are relegated to throwing overhand using either a stiff-elbowed "dart-throwing" motion or a straight-armed motion similar to a cricket bowler, severely limiting their velocity and accuracy.[36] Similarly, the narrow, flat glenoid fossa and robust skeletal architecture of the Neanderthal strongly suggests an inability to efficiently execute the extreme external rotation and rapid kinetic release required for elite projectile throwing.[32] While their immense upper-body strength would have made them exceptional at pushing movements (analogous to the modern shot put), they lacked the specialized biomechanics for high-velocity, lightweight projectile launching.[35]
This biomechanical discrepancy profoundly influenced the respective subsistence strategies and disaster survivability of the two species. Equipped with a shoulder designed for mobility, *H. sapiens* mastered the use of throwing spears, atlatls (spear-throwers), and eventually bows.[35] This allowed them to hunt a highly diverse range of prey—from large ungulates to small, fast-moving avians and mammals—from a safe distance.[35] Furthermore, projectile weaponry provided a distinct tactical advantage during inter-species conflicts and defense against apex predators, drastically reducing the risk of lethal personal injury.[35]
Conversely, the Neanderthal shoulder was optimized for raw, localized power.[32] Their hunting strategy relied overwhelmingly on close-quarters ambush tactics, utilizing heavy, thrusting spears to dispatch large megafauna.[35] While highly effective in the densely forested, closed environments of early Pleistocene Europe, this specialized tactic became a severe liability during periods of rapid climatic deterioration.[40] When forests receded into open steppe-tundras, removing the cover required for ambush hunting, Neanderthals struggled to adapt.[40]
The physical toll of these divergent behaviors is permanently recorded in the bone modeling of the upper limbs. Studies utilizing peripheral quantitative computed tomography on modern elite athletes demonstrate that habitual throwing (as seen in cricket bowlers) induces profound bilateral asymmetry, significantly increasing the robusticity and cross-sectional circularity of the dominant humerus to resist extreme torsional strain.[41] In contrast, habitual swimming induces high, symmetrical robusticity across both arms.[41] The cross-sectional geometries of archaic hominin limbs often reflect highly symmetrical, massive loading, aligning with a lifestyle

dominated by heavy thrusting and potential aquatic foraging, whereas the asymmetric loading profiles increasingly present in later *H. sapiens* populations corroborate the habitual, widespread adoption of high-velocity projectile technology.[32]

Solar Radiation, Folate Protection, and Convergent Depigmentation

Survival across varying latitudes during the Pleistocene mandated rigorous physiological adaptation to diverse and fluctuating levels of ultraviolet radiation (UVR). The integumentary system, specifically epidermal skin pigmentation, serves as the primary biological interface between the hominin organism and the external solar environment. The evolution of human skin pigmentation represents a continuous, delicate evolutionary balancing act: shielding essential, light-sensitive biological compounds from UVR-induced destruction while simultaneously permitting sufficient UVR penetration for indispensable endocrine functions.[43]

Melanin, the primary pigment determining skin color, acts as a highly effective, natural biological shield. In tropical geographic regions situated close to the equator, populations experience intense, year-round UVR exposure.[43] In these environments, natural selection heavily favors high concentrations of eumelanin, resulting in permanently darker skin tones.[43] A critical evolutionary driver for this intense pigmentation is the protection of folate, a highly sensitive water-soluble B-vitamin that is absolutely vital for DNA synthesis, cellular division, and reproductive success.[43] Excessive exposure to intense UVR causes the rapid photolysis (destruction) of folate as blood circulates through the dermal capillaries.[43] Folate depletion during pregnancy leads directly to severe developmental anomalies in offspring, most notably fatal neural tube defects, thereby exerting a massive negative selection pressure on lightly pigmented individuals in high-UVR environments.[43]

Conversely, as hominin populations migrated out of the African tropics and expanded into the higher latitudes of Eurasia, they encountered environments characterized by significantly lower, highly seasonal, and oblique UVR levels.[44] In these temperate and sub-arctic zones, maintaining heavy concentrations of melanin became severely maladaptive. The synthesis of Vitamin D—a hormone essential for the intestinal absorption of calcium, skeletal mineralization, and immune system modulation—requires ultraviolet B (UVB) radiation to penetrate the epidermis and catalyze its precursor molecules.[43] In high-latitude regions, heavily pigmented skin blocks the minimal available UVB, precipitating severe Vitamin D deficiency.[43] This deficiency leads to rickets (which can deform the pelvis and cause fatal obstructed labor), osteomalacia, and severe immune suppression. Consequently, a reduction in eumelanin levels, resulting in light skin, evolved as a necessary adaptation to maximize Vitamin D production in low-light environments.[43]

Genomic sequencing of archaic hominins has revealed complex pigmentation histories. Denisovans, an archaic lineage that inhabited vast stretches of Asia, possessed genetic profiles indicating they likely exhibited dark skin, brown eyes, and dark hair.[46] In contrast, genetic analysis of Neanderthal populations from distinct geographical regions (such as Spain and Italy) reveals that they carried a derived, mutated state of the *MC1R* gene.[47] This specific *MC1R* mutation, distinct from any variant found in modern humans, severely reduced melanin production, endowing these Neanderthals with very pale skin and red hair.[47] Crucially, the genetic pathways leading to light skin in Neanderthals and modern European/East Asian populations represent independent cases of convergent evolution.[47] Modern humans did not inherit their primary light-skin alleles from Neanderthals; rather, modern Eurasian populations developed their own distinct gene variants for depigmentation quite recently, approximately 10,000 years ago, driven heavily by the dietary shift away from Vitamin D-rich hunter-gatherer diets toward cereal-based agriculture.[47]

However, the interbreeding between migrating *H. sapiens* and indigenous Neanderthals approximately 100,000 years ago did leave an indelible mark on the modern human genome.[49] Neanderthals had occupied Eurasia for hundreds of thousands of years and were finely attuned to the local solar and pathogenic environments.[49] Interbreeding resulted in the introgression of Neanderthal alleles into the *H. sapiens* gene pool, with non-African populations today retaining approximately 2% Neanderthal DNA.[49] While many introgressed archaic alleles were deleterious and subsequently purged by purifying selection, several specific alleles facilitated rapid human adaptation to the Eurasian environment.[3] Large-scale genome-wide association studies utilizing databases like the UK Biobank have demonstrated that retained Neanderthal DNA significantly influences multiple modern phenotypes linked to sunlight exposure, including variations in skin tone, the ability to tan, hair color, and even circadian rhythms governing sleep patterns.[49] This genetic introgression effectively provided modern humans with a localized evolutionary shortcut, enhancing their survivability in the northern latitudes without requiring them to wait millennia for novel, advantageous de novo mutations to independently arise.

High-Altitude Hypoxia, the EPAS1 Haplotype, and Denisovan Admixture

Just as intense solar radiation acted as a severe selective pressure shaping hominin integumentary physiology, extreme atmospheric environments tested the absolute limits of migrating populations. The colonization of high-altitude regions, specifically the vast Tibetan Plateau situated over 4,000 meters above sea level, presents one of the most extreme environmental challenges to hominin biology due to severe hypoxia.[52] In these alpine environments, atmospheric oxygen pressure is approximately 40% lower than at sea level.[52] Unacclimated, lowland-adapted modern humans ascending to such altitudes suffer from chronic mountain sickness, marked by dangerous increases in infant mortality and severe reproductive complications.[52] These pathologies are primarily driven by polycythemia—an extreme overproduction of red blood cells triggered by oxygen deprivation, which dangerously increases blood viscosity, impeding microcirculation and straining the cardiovascular system.[52] Despite these hostile conditions, indigenous modern human populations, such as Tibetans and Sherpas, have successfully and permanently settled the plateau.[52] This profound example of environmental survivability is mediated by one of the most remarkable instances of adaptive introgression in human evolutionary history. High-resolution genomic scans have identified that Tibetan populations possess a highly unusual, 32.7-kilobase haplotype located within the *EPAS1* gene (Endothelial PAS domain-containing protein 1).[52] The *EPAS1* gene functions as a critical transcription factor regulating the body's physiological response to hypoxia, operating specifically within the hypoxia-inducible factor (HIF) biological pathway.[4]

Sequencing data confirms that this specific, altitude-adapted *EPAS1* haplotype was not the result of a slow, incremental mutation process occurring solely within the *H. sapiens* lineage. Instead, it was acquired directly through admixture with Denisovans.[52] The core Denisovan haplotype (characterized by the AGGAA motif) is found at extraordinarily high frequencies in Tibetans but is nearly absent or found at highly diminished frequencies (less than 5%) in lowland Han Chinese populations.[52] The Denisovan-derived alleles for *EPAS1*, specifically interacting with regulatory variants such as rs370299814 and the derived G allele of rs150877473, function to fundamentally alter the gene's expression.[53] Rather than triggering a massive overproduction of red blood cells, this introgressed sequence down-regulates *EPAS1* expression in the placenta and endothelial cells during hypoxic stress, effectively blunting the polycythemic response and protecting the organism from cardiovascular failure at high altitudes.[52]

The archaeological record corroborates this genomic data. Recent discoveries of Denisovan fossil material and DNA recovered from karst caves situated at 3,280 meters elevation on the Tibetan Plateau provide definitive evidence that this enigmatic archaic group inhabited the high-altitude region during the Middle Pleistocene.[55] They had successfully adapted to life in a hypoxic environment hundreds of thousands of years before the arrival of anatomically modern humans.[55]

This adaptive introgression acted as a massive evolutionary catalyst for *H. sapiens*. Genomic modeling indicates that an East Asian-specific Denisovan introgression event introduced this highly beneficial sequence into the ancestral population of modern Tibetans.[4] Intriguingly, the sequence remained selectively neutral within the broader lowland population for an extended period.[4] However, concurrent with the permanent inhabitation of the Tibetan Plateau following the retreat of the Last Glacial Maximum (LGM), this archaic haplotype underwent one of the strongest selective sweeps ever recorded in the human genome.[4] Without this specific genetic inheritance from their archaic Denisovan cousins, the modern human expansion into, and survival within, the high-altitude ecological refugia of the Himalayas would have been fundamentally, perhaps insurmountably, compromised.

Denisovan Postcranial Morphology and the Jinniushan Assemblage

While the genomic legacy of the Denisovans in modern humans is well documented through traits like the *EPAS1* adaptation, their physical morphology has long remained obscured by a remarkably sparse fossil record. Until recently, Denisovan paleontology was restricted to highly fragmentary remains recovered from the Denisova Cave in Siberia, primarily consisting of isolated molars and a single, morphologically uninformative proximal fragment of a fifth finger phalanx (Denisova 3).[56] The massive Denisovan molars display archaic characteristics closely aligning with *Homo erectus*, lacking the derived features seen in modern humans.[56]

However, advancements in ancient DNA analysis and morphological comparative studies have begun to illuminate the Denisovan postcranial skeleton. Researchers have definitively linked a distal fragment of a fifth finger phalanx from Denisova Cave as the missing counterpart to the original Denisova 3 specimen.[56] Morphometric analysis of this complete phalanx reveals that its dimensions and structural shape fall entirely within the standard variability of modern *Homo sapiens*, and are distinctly separate from the robust, hyper-specialized fifth finger phalanges of Neanderthals.[56] This indicates that, unlike their archaic dentition, the Denisovan manual postcranial bone structure was plesiomorphic, sharing highly conserved, gracile traits with modern humans.[56]

Furthermore, paleoanthropological focus has shifted toward the Middle Pleistocene hominin fossils recovered from Jinniushan in northeastern China. Dated to approximately 260,000 (±20,000) years ago, the Jinniushan assemblage is exceptionally rare because it contains abundant, clearly associated cranial and postcranial elements belonging to a single adult female individual.[58] Because the Jinniushan specimen shares significant craniodental affinities with other regional fossils (such as *Homo longi* and the Xujiayao hominin) that are increasingly grouped under the Denisovan umbrella, it provides a critical window into Denisovan body morphology.[58]

The Jinniushan fossil elements are highly comprehensive, including a partial cranium, six vertebrae (one cervical and five thoracic), ribs, a complete left patella, a complete left ulna, articulated bones of the hands and feet, and a complete left os coxae (pelvis).[58] The analysis of these postcranial remains, particularly the wide pelvic inlet structure and the robusticity of the long bones, reveals an archaic hominin that was powerfully built, retaining primitive traits adapted for high mechanical loading, yet distinct from the specific cold-adapted, hyper-robust bauplan of the classic European Neanderthal.[59] The integration of the Jinniushan data suggests that Denisovans possessed a mosaic anatomy—combining highly archaic cranial and dental features with a postcranial skeleton that shared functional affinities with both robust archaic forms and certain plesiomorphic traits maintained in modern humans, enabling them to exploit a massive geographic range stretching from the Siberian tundra to the tropical environments of Southeast Asia.[60]

Bioenergetics, Thermoregulation, and Caloric Economics

Perhaps the most restrictive variable governing hominin survivability during the erratic climatic oscillations of the Pleistocene—particularly during Marine Isotope Stage 3 (MIS 3)—was caloric economics. The survival of any species is fundamentally dictated by the precarious balance between daily energy expenditure (DEE) and the energetic return harvested from the surrounding environment.[5] The massive, hyper-arctic body form of the Neanderthal imposed staggering metabolic demands that ultimately rendered them highly vulnerable to resource depletion during ecological downturns.

Adhering strictly to Bergmann's rule of thermoregulation, Neanderthals evolved capacious, barrel-shaped chests and a massive overall body mass to minimize their surface-area-to-volume ratio, an adaptation that highly effectively reduced heat loss in the brutal glacial climates of Pleistocene Europe.[24] However, this cold-adapted morphology, coupled with the extremely high physical activity levels required for close-quarters hunting, necessitated an immense basal metabolic rate (BMR).[62] Predictive bioenergetic models, utilizing regressions based on estimated body mass, fat-free mass, and extreme "hyper-arctic" environmental variables, calculate that female Neanderthals required a baseline BMR of approximately 1,435 kilocalories per day, while males required a baseline BMR of 1,841 kilocalories per day merely to sustain essential resting bodily functions.[64]

When the energetic costs of locomotion, thermoregulation in sub-freezing temperatures, and intense physical exertion are added via the additive energetic method, the total Daily Energy Expenditure for Neanderthals skyrockets.[63] Anthropological models estimate that an average adult Neanderthal required a DEE ranging from 3,100 to 4,500 kilocalories per day.[6] The burden was most severe on reproducing females; a pregnant or lactating Neanderthal female faced extreme metabolic pressures, requiring an estimated 5,500 kilocalories daily—roughly the caloric equivalent of consuming ten modern fast-food hamburgers every single day just to survive.[6]

This immense caloric threshold enforced a highly rigid, high-risk subsistence strategy. Carbon and nitrogen isotopic analyses, alongside extensive faunal profiles, indicate that Neanderthals functioned as apex predators, relying almost exclusively on the targeted hunting of large-bodied game.[65] Their diet exhibited a remarkably narrow breadth, with very little nutritional contribution derived from lower-ranked resources such as plants, small fast-reproducing game, or aquatic foods during inland occupations.[65] In a stable environment where encounter rates with highly ranked prey (such as mammoth, woolly rhinoceros, auroch, or bison) were high, this strategy was optimal, as a single large kill provided a massive, concentrated caloric return.[65] However, when rapid climatic oscillations triggered ecological turnover and depressed these large game populations, Neanderthals were trapped by their own metabolism.

Fossil evidence reflects this immense nutritional strain. Unsurprisingly, Neanderthal dentition frequently preserves "Harris lines"—distinct pathologies in enamel formation that indicate periods of severe, prolonged nutritional stress and starvation during childhood development.[6] Furthermore, physiological constraints limited their dietary flexibility. Once protein consumption exceeds approximately 35% of total caloric intake, the hominin liver and kidneys cannot clear the toxic urea byproduct fast enough, leading to rapid organ damage—a condition of particular danger to pregnant women.[67] While the wide lower thorax of the Neanderthal suggests they may have possessed enlarged livers, slightly increasing their tolerance for a hyper-carnivorous, protein-heavy diet, they still required significant fat yields from their prey to survive.[67] During periods of starvation, attempting to survive by hunting highly lean, small game (such as roe deer or hare, which offer minimal caloric returns) or resorting to cannibalism (where an entire hominin yields fewer calories than a single large herbivore, yet requires massive energetic output and risk to hunt) was mathematically unsustainable.[66]

In stark contrast, dental microwear texture analysis reveals that while Neanderthals constantly altered their diets based strictly on immediate, localized fluctuations in food availability, modern

Upper Paleolithic humans maintained a highly stable diet that was heavily insulated against slight vegetation and climatic changes.[5] This stability was achieved not through a superior metabolism, but through the deployment of advanced technological complexes and broadened resource exploitation strategies that fundamentally shifted their bioenergetic math, allowing them to extract calories more efficiently from a wider array of flora and fauna.[5]

Locomotor Energetics: Endurance Running Versus Sprint Kinetics

In the context of terrestrial movement across wildly fluctuating environments, the baseline energy costs of locomotion significantly influence a species' viable foraging radius, their applicable hunting techniques, and their capacity to rapidly migrate away from natural disasters. Comparative biomechanical and metabolic analyses reveal highly distinct locomotor profiles for modern humans versus archaic hominins like the Neanderthal.

Neanderthals possessed an appendicular skeleton specifically adapted to the cold, heavily forested environments of Middle Pleistocene Europe. They exhibited markedly shortened distal limb segments—specifically the tibia and fibula (the bones below the knee)—resulting in a highly compact, stocky stature.[40] While this conforms perfectly to Allen's rule for heat retention via reduced surface area, it fundamentally altered their gait kinetics.[40] Shorter lower limbs, combined with a highly robust, heavily muscled skeletal frame, optimize the biological leverage and torque required for short, explosive bursts of speed.[40] Consequently, Neanderthals were biomechanically engineered for sprinting, an adaptation heavily favored for the ambush hunting required to take down large fauna in densely wooded terrain.[40]

Conversely, the anatomical configuration of *H. sapiens* reflects a profound evolutionary commitment to endurance running (ER). Endurance running—defined as the ability to sustain aerobic metabolism while running distances exceeding 5 kilometers—is a uniquely derived trait among primates.[39] Originally emerging in *Homo erectus* approximately 2 million years ago as an adaptation for persistence hunting in open savanna environments, this capability is facilitated by a suite of highly specialized physiological features in modern humans.[39] These include the development of the longitudinal arch of the foot and the presence of long, spring-like tendons, such as the Achilles tendon, which store elastic strain energy upon impact and release it during propulsion, effectively reducing the metabolic cost of running by up to half.[39]

A critical skeletal correlate used to measure running economy in the fossil record is the length of the Achilles tendon moment arm, which can be accurately approximated by measuring the length of the calcaneal tuber (the heel bone).[70] Shorter moment arms permit significantly greater storage and release of elastic energy, thereby reducing metabolic costs.[70] Skeletal analyses demonstrate conclusively that Neanderthals possessed relatively longer calcaneal tubers compared to contemporaneous early *H. sapiens* and extant modern humans.[70] This anatomical difference indicates that the energy costs of endurance running would have been substantially higher, and therefore highly inefficient, for Neanderthals.[70] Their inability to efficiently run down prey over long distances limited their ecological flexibility, permanently tying them to ambush tactics that suffered catastrophic failure rates when deforestation and

expanding open steppe-tundras drastically altered the European landscape.[40]
However, it is vital to rigorously contextualize human locomotor efficiency within the broader macroecological scaling laws of mammals. Debates surrounding hominin bipedality frequently overestimate the uniqueness of human energetics. To address this, physiological studies have calculated the net cost of transport (NCOT)—the metabolic rate divided by speed, measured in oxygen consumption per meter—across 81 distinct mammalian species.[71] The comparative models demonstrate significant phylogenetic signaling in both log-transformed body mass and log-transformed NCOT.[71]
When establishing a predictive regression line for NCOT based on mammalian body mass, human walking proves to be highly efficient, requiring approximately 25% less oxygen than predicted for a generic mammal of identical mass.[71] Paradoxically, human running requires 27% *more* oxygen than predicted.[71] Crucially, while modern humans are efficient walkers, all bipedal hominin data points (including estimates for early *Australopithecus afarensis*) fall comfortably within the 95% prediction interval for all mammals.[71] This indicates that, statistically speaking, human bipedal energetics are not fundamentally exceptional or rule-breaking when corrected for phylogenetic history and body mass.[71] Therefore, the survival advantage of *H. sapiens* did not stem from an outright defiance of mammalian metabolic laws, but rather from the specific, targeted application of endurance biomechanics to exploit open-landscape persistence hunting—a highly lucrative ecological niche that was physically unavailable to the sprint-adapted, ambush-dependent Neanderthal.

Technological Buffering: Microclimatic Clothing and the Eyed Needle

A primary differentiator in the bioenergetic equation between the two species was the modern human implementation of sophisticated thermal technology, primarily in the form of advanced, tailored clothing. For decades, some anthropologists hypothesized that the cold-adapted morphology of Neanderthals precluded the necessity for clothing.[63] However, robust ethnographic predictive models—utilizing vast databases incorporating 595 individuals from 245 modern hunter-gatherer groups, cross-referenced with National Oceanic and Atmospheric Administration wind chill indices—have conclusively determined that clothing was an absolute, biological imperative for Neanderthal survival in Pleistocene Europe.[63]
Despite this necessity, the structural complexity of Neanderthal garments remained primitive. Based on paleoenvironmental reconstructions, the analysis of specific faunal remains, and comparisons with traditional societies, researchers deduce that Neanderthals utilized relatively simple, un-tailored garments, functionally akin to draping, pliable ponchos or capes.[63] While they possessed awls and burins to pierce hides, their garments lacked the structural integrity required to completely seal off the body from extreme wind and sub-zero temperatures.[74]
In contrast, *H. sapiens* introduced a revolutionary, paradigm-shifting technological innovation into the archaeological record: the eyed sewing needle.[75] The advent of the eyed needle allowed modern humans to transcend merely wrapping themselves in skins. It enabled the construction of complex, multi-layered, heavily tailored, and form-fitting garments.[74] This

specialized cold-weather clothing effectively sealed off the body, creating a highly stable, artificial microclimate directly against the skin.[74]

By technologically mitigating the external cold, modern humans drastically lowered their somatic thermoregulatory costs. They effectively outsourced their thermal regulation from their internal metabolism to their external material culture. This massive reduction in daily caloric expenditure translated directly into lower infant mortality rates, longer lifespans, and higher overall population survivorship.[5] Furthermore, because they were not expending thousands of calories merely to stay warm, modern humans could afford to exploit a broader, more reliable array of smaller game and plant foods, ensuring demographic stability even as the overarching macroclimate severely deteriorated. The eyed needle also facilitated the adornment of clothing for social and cultural expression, shifting the paradigm of garments from mere biological protection to complex expressions of identity and group affiliation, further strengthening human social bonds.[75]

Demographic Resilience, Social Connectivity, and the Extinction Calculus

The myriad physical, biomechanical, genetic, and technological advantages possessed by *H. sapiens* ultimately funneled into the most critical, overarching determinant of species survivability: demographic resilience and social network connectivity. A landmark 2026 digital ecology study, led by the Université de Montréal, definitively concluded that Neanderthals did not go extinct merely due to a sudden climate shift or direct violent confrontation with modern humans. Rather, the decisive, fatal factor in their disappearance was the profound fragility of their social connectivity.[1]

The high caloric demands and specific ambush-hunting requirements of the robust Neanderthal severely limited the biological carrying capacity of the European environment. Consequently, Neanderthals lived in highly fragmented, small, and isolated bands, averaging approximately just 20 individuals per group.[7] Genomic analyses of Neanderthal remains recovered from caves in the Altai Mountains of Siberia reveal stark, undeniable indicators of massive inbreeding and perilously low genetic diversity within these familial units.[7] In isolated, small populations, genetic drift rapidly and indiscriminately eliminates beneficial gene variants.[7] In a 20-person band, a single hunting accident or localized disease outbreak can permanently erase crucial alleles from the local gene pool.[7]

While archaeological evidence confirms that Neanderthals did maintain some rudimentary regional connections—evidenced by the localized movement of physical lithic materials across the landscape—these links were highly fragile and strictly regionally limited.[1] When the rapid, unpredictable climate variability of the Last Glacial Cycle struck—characterized by violent shifts between freezing stadial phases and warmer interstadial phases—this brittle social architecture fractured entirely.[1] Eastern European Neanderthal populations suffered from extreme, fatal isolation as environments worsened, completely cut off from demographic rescue.[1] Western populations on the Iberian Peninsula survived slightly longer due to marginally more stable core areas, but they too eventually succumbed as their isolated networks failed to provide the support necessary to survive environmental shocks.[1]

Modern *H. sapiens*, conversely, instinctively architected highly complex, expansive, and resilient social networks. Genomic sequencing of anatomically modern humans from the Upper Paleolithic—such as the remains from the 34,000-year-old Sunghir site in Russia—reveals that these early humans recognized the severe biological dangers of inbreeding.[2] They purposefully and systematically sought mating partners far beyond their immediate familial units.[2] This outbreeding was facilitated by the development of sophisticated mating systems, foreshadowing modern marriage ceremonies, complete with complex rituals, symbolism, and the exchange of jewelry and tools to cement inter-group alliances.[2]

This cultural practice generated vast, highly interconnected social networks spanning enormous geographical expanses. The digital ecology study demonstrated that a typical *H. sapiens* local group of 25 to 50 individuals required an annual territory of roughly 2,500 square kilometers to maintain these regional connections and move seasonally.[1] During periods of extreme, rapid climate variability—which the study identified as a far deadlier threat to ancient populations than mere drops in average temperature—these highly connected networks functioned as an indispensable biological and cultural safety net.[1] When localized resources failed, human bands could leverage these alliances to move into neighboring territories, forge temporary cooperative hunting partnerships, and pool vital, life-saving information regarding shifting animal migrations and water resources.[1]

The demographic interaction between these two species was highly complex. The out-of-Africa migration of modern humans resulted in extremely low, yet highly impactful, rates of interbreeding. Mathematical modeling indicates that the genetic admixture required to leave a 2% Neanderthal footprint in modern genomes could have been achieved by an incredibly low rate of gene flow—equivalent to the exchange of a single pair of individuals between the two populations roughly every 77 generations.[79] This incredibly sparse rate of interbreeding likely explains the complete absence of Neanderthal mitochondrial DNA in modern humans.[79] Furthermore, genomic evidence suggests that the original Neanderthal Y chromosome was entirely replaced by an extinct lineage of the modern human Y chromosome via introgression between 100,000 and 370,000 years ago, highlighting the severe genetic load and demographic weakness inherent in the declining Neanderthal population.[79] As *H. sapiens* expanded into Western Europe, their robust demographic networks and ability to extract higher caloric yields from the environment placed insurmountable demographic stress on the already vulnerable, highly isolated Neanderthals, ultimately absorbing their remnants and precipitating their total extinction.[1]

Conclusions

The triumph of *Homo sapiens* and the concomitant extinction of the Neanderthal and Denisovan lineages cannot be distilled into a single, simplistic variable of cognitive or martial superiority. The survival of modern humans through the catastrophic environmental volatility of the Pleistocene was the result of a highly complex, compounding sequence of eco-physiological adaptations, biomechanical versatility, and socio-cultural innovations that created a fundamentally unassailable survivability matrix.

Neanderthals and archaic hominins were extraordinary organisms, perfectly, yet rigidly, engineered for a specific ecological paradigm. Their robust, pachyosteosclerotic skeletons provided essential hydrostatic ballast, allowing them to exploit shallow aquatic resources and withstand massive physical trauma. Their specific shoulder architecture and shortened distal limbs made them unparalleled, explosive ambush predators in the dense forests of early Europe. However, these extreme morphological specializations carried lethal energetic costs. The immense daily caloric requirement necessary to maintain their hyper-arctic body mass and baseline metabolic rate forced a strict reliance on high-risk, large-game hunting. This severe caloric bottleneck artificially capped their population densities, leading invariably to small, isolated, and genetically stagnant bands highly susceptible to rapid environmental shifts and genetic drift.

Modern *Homo sapiens* evolved a fundamentally divergent paradigm based on profound systemic flexibility. A gracile skeleton characterized by low trabecular density afforded dynamic buoyancy and superior locomotor efficiency, drastically lowering the basal physiological cost of survival. Complex biomechanical adaptations within the shoulder complex unleashed the kinetic power of the high-velocity projectile, fundamentally altering the risk-to-reward ratio of hunting and defense. Adaptive genetic introgression from archaic populations provided immediate, critical physiological upgrades, seamlessly equipping modern humans with vital protections against intense solar radiation and high-altitude hypoxia. Crucially, technological innovations, particularly the advent of heavily tailored, microclimatic clothing facilitated by the eyed needle, effectively decoupled human thermoregulation from raw, internal metabolic heat generation.

Ultimately, this massively reduced caloric footprint permitted the sustainment of much higher population densities. These demographics, in turn, fueled the intentional creation of expansive, deeply interconnected social networks and outbreeding alliances spanning thousands of kilometers. When the violent, unpredictable climatic oscillations of the Last Glacial Cycle devastated localized ecologies, the heavily specialized Neanderthals died in isolated fragmentation. *Homo sapiens* survived because their morphological and metabolic efficiency built the cultural and demographic safety nets necessary to weather the storm, effectively engineering a resilience that far transcended the biological limits of the individual organism.

Works cited

1. Scientists think they finally know why Neanderthals vanished ..., accessed May 29, 2026, https://www.sciencedaily.com/releases/2026/04/260427050609.htm
2. Prehistoric humans are likely to have formed mating networks to avoid inbreeding, accessed May 29, 2026, https://www.cam.ac.uk/research/news/prehistoric-humans-are-likely-to-have-formed-mating-networks-to-avoid-inbreeding
3. The Contribution of Neanderthal Introgression to Modern Human Traits - PMC - NIH, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9741939/
4. The history and evolution of the Denisovan-EPAS1 haplotype in Tibetans | PNAS, accessed May 29, 2026, https://www.pnas.org/doi/10.1073/pnas.2020803118
5. Energetic Competition Between Neandertals and Anatomically Modern Humans -

Paleoanthropology Society, accessed May 29, 2026, https://paleoanthro.org/media/journal/content/PA20090096.pdf
6. Neanderthal Country (Chapter 8) - The Unstoppable Human Species, accessed May 29, 2026, https://www.cambridge.org/core/books/unstoppable-human-species/neanderthal-country/5E3B501EBBA16E153D05FF2DE703808C
7. Neanderthals vs Homo sapiens: How social structures affected ancient species' ability to survive - Genetic Literacy Project, accessed May 29, 2026, https://geneticliteracyproject.org/2024/07/30/neanderthals-vs-homo-sapiens-how-social-structures-affected-ancient-species-ability-to-survive/
8. Recent origin of low trabecular bone density in modern humans - PMC, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4299206/
9. On the "Lightness" of Sapiens: Humans did not inherit the robust ..., accessed May 29, 2026, https://www.ibe.upf-csic.es/news/-/asset_publisher/PXTgqZXxlocA/content/on-the-lightness-of-sapiens-humans-did-not-inherit-the-robust-bones-of-our-ancestors/maximized
10. Recent origin of low trabecular bone density in modern humans - American Museum of Natural History, accessed May 29, 2026, https://www.amnh.org/content/download/427934/6201470/file/recent-origin-of-low-trabecular-bone-density-in-modern-humans.pdf
11. Pachyosteosclerosis in Archaic Homo: Heavy Skulls for Diving, Heavy Legs for Wading?, accessed May 29, 2026, https://www.eurekaselect.com/chapter/774
12. CHAPTER 5 - Semi-Aquatic Human Ancestors, accessed May 29, 2026, https://aquatic-human-ancestor.org/files/Pachyosteosclerosis_in_Archaic_Homo_Heav.pdf
13. Pachyosteosclerosis Suggests Archaic Homo Frequently Collected Sessile Littoral Foods - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/21741646/
14. Sink or swim? Bone density as a mechanism for buoyancy control in early cetaceans, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/17516430/
15. Correlation of Bone Density in Aquatic and Semiaquatic Animals to Ecological and Dietary Specializations | Crimsonpublishers.com, accessed May 29, 2026, https://crimsonpublishers.com/oara/fulltext/OARA.000537.php
16. Neanderthal skeletons were still pachy-osteo-sclerotic | PLOS One, accessed May 29, 2026, https://journals.plos.org/plosone/article/comment?id=10.1371/annotation/dcc00e58-5850-48f1-9d91-f3fd3b9399b8
17. Pachyosteosclerosis in Archaic Homo: Heavy Skulls for Diving, Heavy Legs for Wading?, accessed May 29, 2026, https://www.researchgate.net/publication/271446186_Pachyosteosclerosis_in_Archaic_Homo_Heavy_Skulls_for_Diving_Heavy_Legs_for_Wading
18. Were Neanderthals Getting Surfer's Ear From Diving for Seafood? - Smithsonian Magazine, accessed May 29, 2026, https://www.smithsonianmag.com/smart-news/neanderthals-had-lots-surfers-ea

r-suggesting-they-were-seafood-180972917/
19. Aural exostoses (surfer's ear) provide vital fossil evidence of an aquatic phase in Man's early evolution - PMC, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5696936/
20. Neanderthals and Ancient Humans Suffered from Swimmer's Ear: Study - Sci.News, accessed May 29, 2026, https://www.sci.news/othersciences/anthropology/neanderthals-ancient-humans-swimmers-ear-07497.html
21. Bones Reveal That Neanderthals Were Prone to Surfer's Ear, But Nobody Knows Why, accessed May 29, 2026, https://www.sciencealert.com/bones-reveal-that-neanderthals-were-also-prone-to-surfer-s-ear
22. External auditory exostoses in Shanidar 1 caused by semi-aquatic lifestyle? | PLOS One, accessed May 29, 2026, https://journals.plos.org/plosone/article/comment?id=10.1371/annotation/4c045aa4-89c3-4573-956b-3f956df45b05
23. Certain Neanderthal skulls show signs of Surfer's ear, which are bone growths formed by the ear caused by exposure to moist environments. suggesting that Neanderthals were diving underwater, possibly for food, foraging or leisure time. : r/Paleontology - Reddit, accessed May 29, 2026, https://www.reddit.com/r/Paleontology/comments/wkvmd8/certain_neanderthal_skulls_show_signs_of_surfers/
24. Neanderthal anatomy - Wikipedia, accessed May 29, 2026, https://en.wikipedia.org/wiki/Neanderthal_anatomy
25. New study shatters heavy bones myth - RNLI, accessed May 29, 2026, https://rnli.org/news-and-media/2025/december/10/new-study-shatters-heavy-bones-myth
26. Swimming Science Bulletin - San Diego State University, accessed May 29, 2026, https://coachsci.sdsu.edu/swim/bullets/float36.htm
27. Clavicle morphology and Neandertal shoulder architecture - ResearchGate, accessed May 29, 2026, https://www.researchgate.net/publication/323615509_Clavicle_morphology_and_Neandertal_shoulder_architecture
28. Scapular Position and Orientation in Throwing Athletes - UK College of Health Sciences, accessed May 29, 2026, https://chs.uky.edu/sites/default/files/2023-12/AJSM_Scapular%20Position%20and%20Orientation%20in%20Throwing%20Athletes.pdf
29. Clavicle length and shoulder breadth in hominoid evolution - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/36594678/
30. Neandertal clavicle length - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/24616525/
31. The Pectoral Girdle (Chapter 6) - Neandertal Skeletal Anatomy, accessed May 29, 2026, https://www.cambridge.org/core/books/neandertal-skeletal-anatomy/pectoral-girdle/24EF50CD68E3C84189FC5EB8C6E4C581

32. Neandertal scapular glenoid morphology - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/2248374/
33. Functional Morphology of the Neandertal Scapular Glenoid Fossa - ResearchGate, accessed May 29, 2026, https://www.researchgate.net/publication/267739873_Functional_Morphology_of_the_Neandertal_Scapular_Glenoid_Fossa
34. The morphology and evolutionary history of the glenohumeral joint of hominoids: A review, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6342098/
35. If Neanderthals still existed today, and they were allowed into the Olympics, is it likely they would beat us in any sports? Which ones? : r/AskBiology - Reddit, accessed May 29, 2026, https://www.reddit.com/r/AskBiology/comments/1rripzn/if_neanderthals_still_existed_today_and_they_were/
36. Hard and Fast | Harvard Medical School, accessed May 29, 2026, https://hms.harvard.edu/news/hard-fast
37. The Evolution of Our Throwing Shoulder - YouTube, accessed May 29, 2026, https://www.youtube.com/watch?v=1y__4xX8xp8
38. Throwing, the Shoulder, and Human Evolution, accessed May 29, 2026, https://cdn.mdedge.com/files/s3fs-public/issues/articles/media_463f64009d0ef984b5_ajo045030110.PDF
39. Humans Were Born to Run, Fossil Study Suggests | National Geographic, accessed May 29, 2026, https://www.nationalgeographic.com/science/article/humans-were-born-to-run-fossil-study-suggests
40. Considering Neanderthal Locomotion – SAPIENS, accessed May 29, 2026, https://www.sapiens.org/biology/neanderthal-locomotion/
41. Habitual throwing and swimming correspond with upper limb diaphyseal strength and shape in modern human athletes - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/19358297/
42. Habitual Throwing and Swimming Correspond With Upper Limb Diaphyseal Strength and Shape in Modern Human Athletes - ResearchGate, accessed May 29, 2026, https://www.researchgate.net/publication/24268372_Habitual_Throwing_and_Swimming_Correspond_With_Upper_Limb_Diaphyseal_Strength_and_Shape_in_Modern_Human_Athletes
43. Did Neanderthals affect our Skin Color? - YouTube, accessed May 29, 2026, https://www.youtube.com/watch?v=Wqp-LbeC344
44. Modern life may cause sun exposure, skin pigmentation mismatch - ScienceDaily, accessed May 29, 2026, https://www.sciencedaily.com/releases/2013/02/130217084323.htm
45. Light skin - Wikipedia, accessed May 29, 2026, https://en.wikipedia.org/wiki/Light_skin
46. accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10250142/#:~:text=For%20example%2C

%20SNVs%20in%20pigmentation,sepsis%20in%20humans%20(95).
47. Studies suggesting Neanderthals and Denisovans were dark-skinned - Reddit, accessed May 29, 2026, https://www.reddit.com/r/pleistocene/comments/1r7657d/studies_suggesting_neanderthals_and_denisovans/
48. Adaptation of human skin color in various populations - PMC, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5502412/
49. How easily we tan is influenced by Neandertal DNA - Max-Planck-Gesellschaft, accessed May 29, 2026, https://www.mpg.de/11533845/neandertal-dna
50. Ancient DNA and Neanderthals - The Smithsonian's Human Origins Program, accessed May 29, 2026, https://humanorigins.si.edu/evidence/genetics/ancient-dna-and-neanderthals
51. The evolution of skin pigmentation-associated variation in West Eurasia | PNAS, accessed May 29, 2026, https://www.pnas.org/doi/10.1073/pnas.2009227118
52. Altitude adaptation in Tibet caused by introgression of Denisovan-like DNA - PMC, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4134395/
53. Wide distribution and altitude correlation of an archaic high-altitude-adaptive EPAS1 haplotype in the Himalayas - PMC, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4796332/
54. Down-Regulation of EPAS1 Transcription and Genetic Adaptation of Tibetans to High-Altitude Hypoxia | Molecular Biology and Evolution | Oxford Academic, accessed May 29, 2026, https://academic.oup.com/mbe/article/34/4/818/2918109
55. Introgressive Hybridization and Hypoxia Adaptation in High-Altitude Vertebrates - Frontiers, accessed May 29, 2026, https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2021.696484/full
56. Morphology of the Denisovan phalanx closer to modern humans than to Neanderthals, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6726440/
57. Morphology of the Denisovan phalanx closer to modern humans than to Neanderthals, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/31517046/
58. Denisovan - Wikipedia, accessed May 29, 2026, https://en.wikipedia.org/wiki/Denisovan
59. The cranium, left os coxae, and left ulna of the Jinniushan specimen.... | Download Scientific Diagram - ResearchGate, accessed May 29, 2026, https://www.researchgate.net/figure/The-cranium-left-os-coxae-and-left-ulna-of-the-Jinniushan-specimen-Note-that-the_fig1_7273053
60. Archaic Hominin Populations in Asia before the Arrival of Modern Humans Their Phylogeny and Implications for the “Southern Denisovans” - The University of Chicago Press: Journals, accessed May 29, 2026, https://www.journals.uchicago.edu/doi/full/10.1086/694318
61. Denisovan Epigenetics Reveals Human Anatomy | The Institute for Creation Research, accessed May 29, 2026, https://www.icr.org/content/denisovan-epigenetics-reveals-human-anatomy
62. Bioenergetic perspectives on Neanderthal thermoregulatory and activity budgets

- Scholars@Duke publication, accessed May 29, 2026, https://scholars.duke.edu/publication/1314086
63. A New Estimate for Neanderthal Energy Expenditure - CUNY Academic Works, accessed May 29, 2026, https://academicworks.cuny.edu/cgi/viewcontent.cgi?article=1350&context=hc_sas_etds
64. Neandertal Cold Adaptation: Technological, Anatomical, and Physiological Responses to Cold Stress in One of Our Closest Fossil Relatives - PMC, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12485224/
65. Plant foods and the dietary ecology of Neanderthals and early modern humans - Smithsonian Research Online, accessed May 29, 2026, https://repository.si.edu/bitstreams/98c7c5df-9438-4176-a2ec-ab550bf371bd/download
66. Assessing the calorific significance of episodes of human cannibalism in the Palaeolithic, accessed May 29, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5382840/
67. Bioenergetic perspectives on Neanderthal thermoregulatory and activity budgets | Request PDF - ResearchGate, accessed May 29, 2026, https://www.researchgate.net/publication/226306274_Bioenergetic_perspectives_on_Neanderthal_thermoregulatory_and_activity_budgets
68. Neandertal versus Modern Human Dietary Responses to Climatic Fluctuations - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/27119336/
69. Sustained Running Behavior | Center for Academic Research and Training in Anthropogeny (CARTA), accessed May 29, 2026, https://carta.anthropogeny.org/moca/topics/sustained-running-behavior
70. Calcaneus length determines running economy: implications for endurance running performance in modern humans and Neandertals - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/21269660/
71. Comparative energetics of mammalian locomotion: humans are not different - PubMed, accessed May 29, 2026, https://pubmed.ncbi.nlm.nih.gov/22963931/
72. Comparative energetics of mammalian locomotion: humans are not different, accessed May 29, 2026, https://research.monash.edu/en/publications/comparative-energetics-of-mammalian-locomotion-humans-are-not-dif/
73. Energy Efficiency Doesn't Explain Human Walking? - Smithsonian Magazine, accessed May 29, 2026, https://www.smithsonianmag.com/science-nature/energy-efficiency-doesnt-explain-human-walking-39161215/
74. How eyed sewing needles facilitated the expansion of early 'Homo sapiens' | Science | EL PAÍS English, accessed May 29, 2026, https://english.elpais.com/science-tech/2024-07-01/how-eyed-sewing-needles-facilitated-the-expansion-of-early-homo-sapiens.html
75. Eyed Needles Were Invented around 40,000 Years Ago in Eastern Eurasia, Archaeologists Say | Sci.News, accessed May 29, 2026, https://www.sci.news/archaeology/eyed-needle-invention-13057.html

76. The beginnings of fashion | ScienceDaily, accessed May 29, 2026, https://www.sciencedaily.com/releases/2024/06/240628160129.htm
77. Homo Sapiens Outlasted Neanderthals Because of Stronger Social Networks, a 2026 Study Finds. Games May Have Been Part of Why - Popular Archeology, accessed May 29, 2026, https://popular-archaeology.com/article/homo-sapiens-outlasted-neanderthals-because-of-stronger-social-networks-a-2026-study-finds-games-may-have-been-part-of-why/
78. Neanderthal genome shows evidence of early human interbreeding, inbreeding, accessed May 29, 2026, https://news.berkeley.edu/2013/12/18/neanderthal-genome-shows-evidence-of-early-human-interbreeding-inbreeding/
79. Interbreeding between archaic and modern humans - Wikipedia, accessed May 29, 2026, https://en.wikipedia.org/wiki/Interbreeding_between_archaic_and_modern_humans